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HARTING's Technology Newsletter

Guest article:

Prof. Dr. Jay Lee

Industry 4.0 □

Factory in Big Data Environment

Leading article:

Claus Hilger

Orchestrating

services

Guest article:

Prof. Dr. Dr. h.c. Detlef Zühlke

□ Industrie 4.0 □ □

from vision to reality

Integrated Industry □ Orchestrating services



Pushing Performance



HQT center
opening on
27 June 2014

HARTING quality

in a new space:
our quality and technology center

The new HARTING quality and technology center (HQT Center) stands where company founder Wilhelm Harting once built the first facility in Espelkamp. Starting in April of this year, our products will undergo the most stringent testing in an ultramodern laboratory here in an area covering 3,500 square meters □all so that they can create value for our customers. The result is the foundation for a successful future built on our company's tradition.

Smart Solutions for Integrated Industry.

Integrated Industry is taking off and mastering the next steps, as evidenced by the theme of the Hannover Messe trade fair. But the fact that Integrated Industry, or Industry 4.0, is on everyone's lips still doesn't make the vision any more real for end users. They need to see solutions, something that demonstrates tangible benefit.



□Philip Harting,
Senior Vice President Connectivity & Networks

Dear customers and business partners,

Talk alone will not help □visionary ideas must also be implemented and tested. Those who want Integrated Industry must develop solutions and vigorously put them into practice.

Consequently, we have developed solutions and tested them in real-world applications. I deliberately chose the plural form of the word, since the solution approaches are as multifaceted as the technologies that are employed. In a nutshell, Integrated Industry is the transformation of today's factories into Smart Factories, where the orchestration of services represents the definitive solution for fully modularized automation.

This edition of tec.News features an entire orchestra of our Smart Solutions.

What interests me here is which Smart Solutions generate the greatest user benefits for you, and in which combination. So, let's talk about these specific solutions. Feel free to speak directly with me, or visit us at one of our next trade fairs to directly assess the benefits you will derive from Integrated Industry, right at the source.

Here's hoping you enjoy your reading!

Yours sincerely,

A handwritten signature in blue ink that reads "Philip Harting".



Orchestrating services



Auto-ID integration □ a bridge between worlds

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Orchestrating services

Customer-tailored manufacturing requires a high degree of flexibility in production. In Smart Factories, modular production units are connected together to form executable processes. Here, the orchestration of services, their standardized description, and service-oriented software architecture all play a central role.

□Claus Hilger, Director of HARTING IT System Integration GmbH & Co. KG, HARTING Technology Group, Claus.Hilger@HARTING.com



Since traditional systems are incapable of meeting flexible customer requirements, manufacturing processes must be modularized and assembled for individual processes depending on the situation. This entails the integration of various technologies into a Smart Factory: automation technology, sensors, RFID and, last but not least, information technology.

SMART FACTORY

In the Smart Factory, production is controlled by components and products in a largely decentralized and autonomous manner. Via sensors, objects detect the state of their surroundings, they can be uniquely identified, possess built-in object memory and exchange information with other objects, e.g. regarding their processing state. Through the addition of embedded systems, objects become cyber physical systems (CPS). The result is a maximum of flexibility and the ability to profitably produce even the smallest batch sizes.

” In the Smart Factory, production is controlled by machines, robots, conveyor systems, loading equipment and products, in a largely decentralized and autonomous manner.

The coordination of individual CPS in the Smart Factory takes place in service-oriented architectures (SOA), which have been successfully employed in information technology for years. The basic idea is to encapsulate individual components as services. Here, CPS can function as service provider or service consumer.

ORCHESTRATION

When orchestrating services, modular production units are assembled into executable production processes. For the modeling, various programming approaches such as BPML (Business Process Modeling Language) and BPEL (Business Process Execution Language) are available.

” There is a constantly increasing desire to customize products, as well as for individually tailored manufacturing.

In addition to the orchestration of services, one of the biggest challenges is the formal description of the services and their standardization. System integration requires a reference architecture that enables providers and users to connect individual components with each other with plug & play capability. This will have to be cleverly designed, since directly transferring IT approaches to automation will not be possible. ■

IN BRIEF

- t Increased flexibility in production
- t Small-batch production for individual customer requirements

Industry 4.0 □ Factory in Big Data Environment



Our guest article:



Prof. Dr. Jay
Lee, Ohio
Eminent Scholar,
L.W. Scott Alter

Chair Professor and Director of
NSF I/ UCRC Center for Intelli-
gent Maintenance Systems (IMS)

Ann Kao, PhD Researcher, NSF
I/ UCRC Center for Intelligent
Maintenance Systems (IMS),
University of Cincinnati

After 2013 Hannover Fair took place in Germany, □Industry 4.0□ is becoming a buzzword in manufacturing industry. **Accompanying this new concept, intelligent analytics and cyber physical systems are teaming together to realize a new thinking of production management and factory transformation.** In the past, most of the manufacturing systems are utilizing a centralized control scheme, in which machines are treated independently. Under this scheme, machining data is collected and utilized to assure a normal production process. Also, once the product quality does not meet the requirements, the collected data can serve as clues to find the root-cause. In order

to achieve transparency, the manufacturing industry has to transform itself into predictive manufacturing.

Today, in an Industry 4.0 factory, machines are connected as a collaborative community. Such evolution requires the utilization of innovative prediction tools so that data can be systematically processed into information that can explain the uncertainties and thereby make more □informed□ decisions.

The conceptual framework of predictive manufacturing system starts with data acquisition of the monitored assets. Using appropriate sensor installations, various signals such as vibration, pressure, etc. can be extracted. In addition, historical data can be harvested for further data mining. Communication protocols, such as MTConnect and OPC, can help users to record controller signals. When all the data are aggregated, this amalgamation is called □Big Data□. The transforming agent consists of several components: an integrated platform, predictive analytics and visualization tools. The algorithms found in the Watchdog Agent□ can be categorized into four sections: signal processing and feature extraction, health assessment, performance prediction, and

fault diagnosis. By utilizing visualization tools, health information (such as current condition, remaining useful life, failure mode, etc.) can be effectively conveyed in terms of a radar chart, fault map, risk chart and health degradation curves.

With manufacturing transparency, management then has the right information to determine facility-wide OEE. With the prediction capability, equipment can be managed cost effectively with just-in-time maintenance. Finally, historical health information can be fed back to the equipment designer for closed loop lifecycle redesign. ■

TODAY'S FACTORY				INDUSTRY 4.0 FACTORY	
	Data source	Key attributes	Key technologies	Key attributes	Key technologies
Component	Sensor	Precision	Smart Sensors and Fault Detection	Self-Aware Self-Predict	Degradation Monitoring & Remaining Useful Life Prediction
Machine	Controller	Producibility & Performance (Quality and throughput)	Condition-based Monitoring & Diagnostics	Self-Aware Self-Predict Self-Compare	Up time with Predictive Health Monitoring
Production System	Networked manufacturing system	Productivity & OEE	Lean Operations: Work and Waste Reduction	Self-Configure Self-Maintain Self-Organize	Worry-free Productivity

Comparison of today's factory and an Industry 4.0 factory

INDUSTRIE



from vision to reality

Our guest article:



Prof. Dr. Dr. h.c.
Detlef Zühlke
Scientific
Director,

Innovative Factory Systems
German Research Center for
Artificial Intelligence DFKI,
Kaiserslautern

For about two years now, a new concept has been attracting attention: **Industrie 4.0**. Underlying this is the vision of an imminent fourth industrial revolution, brought on mainly by advances in information and communication technology. Just as our everyday objects such as mobile phones, cameras, cars, etc. are becoming **smart** and networked into the global Internet, a similar development can also be witnessed inside our factories.

Following the paradigm of the Internet-of-Things, all the components that make up our production systems are becoming smart network nodes that can be easily - and largely autonomously - integrated into IP-based networks. The way we nowadays install and operate printers and other devices via plug & play is also beginning to be mirrored in the way production facilities are built, commissioned and operated.

TASKS

But the reality will not be as simple as this comparison might suggest. The end-to-end Ethernet-based **Industrie 4.0** approach will change fundamentally our existing control architectures. In a long-term view, every sensor and actuator at the field level will be a network node in exactly the same way as an ERP system or NC control. As a result, all the informa-

tion in the entire network will become available.

Still many open questions need to be answered. For example, we need standards on several levels so that the factory elements can be put together like Lego bricks. Also, many problems in the area of data security have to be solved before **Industrie 4.0** can be realized. At least over the next few

“The slogan "Industrie 4.0" is on everyone's lips and has generated a great deal of attention, especially in the media. Now, the time has come to transform "Industrie 4.0" into a reality inside factories.

years, the real-time behavior that is so important in industrial controls will not be achieved on lower control levels using Ethernet-based solutions.

Industrie 4.0 gives us a vision of a more flexible and efficient future. To turn this vision into reality, we need just as with any quantum leap in technology the combined efforts of industry and science. The goal is to turn the vision into products, and because an isolated product cannot be successful in a networked world, we need collaboration between innovative companies from a variety of product areas, all working together to make this implementation reality.

SMARTFACTORY

Under the umbrella of the smartfactory^{KL} project, based in Kaiserslautern, Germany, nine partners are working together to build a prototype **Industrie 4.0** production plant. Each partner company amongst others the HARTING Technology Group - will build one system module or furnish cross-application technology.

smartfactory^{KL} defines the required mechanical, electrical and communication standards and will integrate the individual modules to produce a common demonstrator for the 2014 Hannover Messe trade show. This will comprise a first important step in transforming **Industrie 4.0** into reality. ■

Industrie 4.0 HARTING'S COMMITMENT

HARTING is actively participating in the construction of the new smartfactory^{KL} production facility. The company's involvement extends to two areas:

The construction of an "Industrie 4.0" production module that employs cutting-edge automation technologies. This has already seen the integration of a remote Ethernet-based maintenance concept that has access to every module component.

The testing of our infrastructure concepts via the integration of power management. Here, the modules are made uniformly pluggable via Han-Modular[®].

smartfactory^{KL} has provided us with the environment to integrate ourselves into innovative automation systems and to demonstrate their benefits to our customers.

Adaptive process models for flexible production technology

How man and machine learn from each other

Individualized production requires flexible and vertically integrated software and hardware platforms. Modularization and integrated process modeling will enable easy configuration of customized production systems. Interactive learning methods for man and machine assist in the development of the required process models.

□ Dr. Volker Franke, Director of Applied Technologies, HARTING Technology Group, Volker.Franke@HARTING.com

□ Dr.-Ing. Sebastian Wrede, Group Leader Cognitive Systems Engineering, CoR-Lab & CITEC, Bielefeld University



The individualization of production systems based on specific customer requirements calls for flexible manufacturing solutions. HARTING customers can already configure and order customized connectors via the Internet. Future intelligent production systems, so-called Smart Factories, will derive production orders from these order and schedule them autonomously. Production systems will be fully networked, based on digital manufacturing and product data with flexible manufacturing modules. This requires modular manufacturing platforms that can be configured just as efficiently as today's smartphones but which possess the process reliability and effectiveness of modern manufacturing lines.

PROCESS MODELING AS A BASIS FOR FLEXIBLE PRODUCTION

Within the framework of the leading edge cluster it² OWL, HARTING is conducting research in close collaboration with Bielefeld University in the FlexiMon project in order to realize this vision. A core objective is the development of flexible manufacturing cells with **plug-and-produce** capabilities that can be configured with minimal changeover times and integrated vertically into corporate IT via open standards. A crucial component within this concept is a process model that remains consistent at all layers and process steps, which permits a complete description of the production logic.

As part of FlexiMon, the Business Process Modeling Notation (BPMN 2) standard is extended for this purpose and

used to model production processes down to the machine level. Due to the ubiquitousness of BPMN in enterprise IT, it promises a smooth mapping of production logic to the ERP level. A central question, however, is what extensions and model transformations are required to make BPMN models executable on the machine level.

The basic idea is that, should a customer request it, a production process described in BPMN can be set up at

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the ERP level itself and successively augmented with information about the production process. For example, if information is missing for execution at the machine level, an expert in the production process can add it to the process model directly on the machine by way of user friendly interface.

INCREMENTAL OPTIMIZATION THROUGH INTERACTIVE LEARNING

In order to make the risk of errors due to the increased technical complexity more manageable, employees that work with the production system either in configuration or production modes are supported by assistance functions. The man-machine interface allows intuitive online designation of operational conditions and their association with production data and active process steps. This data permits the application of machine

learning methods for parameter and process optimization or for condition monitoring. Eventually, these methods shall facilitate automatic improvement of process models as well as better support for human worker during interactive configuration and operation of the production system. In this way, employees can be involved in more flexible forms of production, as well as actively shape them and contribute to process improvements. ■

➔ IN BRIEF

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Auto-ID integration a bridge between worlds

Integration means connecting different worlds. As HARTING is a specialist in the realization of reliable connections, we feel at home in this universe. But which Auto-ID technology is capable of doing what, and how can this integration look like? Depending on the application, integration can range from simple to complex, from relays all the way through to SAP.

□ Claus Hilger, Director of HARTING IT System Integration GmbH & Co. KG, HARTING Technology Group, Claus.Hilger@HARTING.com

□ Dr. Jan Regtmeier, Product Manager RFID Reader, HARTING Technology Group, Jan.Regtmeier@HARTING.com

When transferring media □ say from paper to IT systems □ much data and information can be lost. As a result, the potential for efficiency gains and cost reduction can fall by the wayside. In the case of production processes under high expectations and cost pressures, however, this is unacceptable.

Auto-ID technologies are capable of minimizing such data loss, but in order to achieve this they need to be sufficiently integrated into both the worlds they connect. Currently, three Auto-ID technologies are deployed: barcode (including QR), image recognition and RFID. Each of these systems has its clear strengths:

” Auto-ID technologies have the potential to minimize data loss during media conversions to the IT world and hence optimize processes.

barcode is extremely favorable in terms of costs and easy to implement. Image recognition does not require any additional information. RFID, however, is the only technology that simultaneously permits unique identification, confers the object with memory, and allows bi-directional communication between the object and the IT world (see table).

The information gained by means of Auto-ID must be prepared in such a way

that the target IT system can process it. Depending on the application, entirely different software components are necessary for this: a PLC function block, middleware, the application suite or the SAP module Auto-ID Infrastructure (AII). The following are some examples of integration, ranging from simple to complex, in which different components are used for integration in each case.

CROSS-INDUSTRY DESIGNATION



OCR



	Barcode	OCR	RFID
Unique ID code	✓	✓	✓
Without printing		✓	
Changeable			✓
Memory			✓
Bi-directional communication			✓
Sensors			✓



Example 4: RFID used for incoming goods

Example 1: Local trains are getting longer and longer, while the platforms are not. Naturally, only doors which open onto a platform should be opened. This can be detected by way of RFID. The RFID reader switches on a relay that shows the train operator which doors he or she can open.

Example 2: Modern machines consist of numerous modules. Executing a PLC program that does not match the current module configuration can cause serious damage to machines. The RFID reader recognizes the modules and communi-

cates with the PLC by means of a function block, thereby ensuring that only authorized programs are executed.

Example 3: Service calls are expensive, which makes trouble-free processing and sequences all the more important. By way of example, a handheld is used to identify a machine listed for servicing. The service technician is led through the process step by step via an app (implemented via the Ha-VIS Application Suite). Each step is simultaneously documented and archived in a database.

Example 4: Today, the receipt of goods should be posted automatically. Using RFID, hundreds of objects can be simultaneously detected and sent directly to the SAP AII module via Ha-VIS Middleware.

Here, **HARTING** offers the most expedient products worldwide, from transponders through to the software packages, as well as additional consulting and integration services in German-speaking regions □all of which makes for expert Auto-ID integration from a single source. ■



Example 1:
RFID in rail traffic and transport



Example 2:
RFID in modern machinery



Example 3:
The Ha-VIS Application Suite simplified maintenance



Ha-VIS mCon 3000

PROFINET switches □

reliable, adaptable and flexible

Stable, secure and reliable data transfer is one of the most critical factors in automation technology. Demands are increasing in parallel with the flexibility of modern systems and the need to reuse components in a modularized system layout. Ethernet and PROFINET are internationally valid standards and support the construction of a cross-vendor communication platform.

□Oliver Opl, Product Manager, HARTING Technology Group, Oliver.Opl@HARTING.com





Manufacturer-specific communication solutions are specially tailored to particular use cases and applications in order to generate specific advantages and effects. In many cases, being able to use these advantages in the field requires significant effort in the selection of components, as well as precise planning □ which is not always possible. Proprietary solutions can also result in additional constraints if they are incompatible with existing components or machines. Alternatively, increased costs can be expected for specially adapted components.

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Consequently, standardized components such as the managed Ethernet switches of the new Ha-VIS mCon 3000 family offer decisive design, performance and cost advantages. They comply with Ethernet standard IEEE 802.3 and also meet all the requirements of PROFINET Conformance Class B (CCB). This guarantees global compatibility with all components that also meet these international standards.

Here, PROFINET combines all the advantages of classic field-bus systems (fast cycle times) and Ethernet (free choice of topology). Guaranteed service quality for data transmission is ensured by methods such as Quality of Service (QoS). This makes it possible to relay critical and control-relevant data packets with high priority and implement applications with very short cycle times. The standard even ensures quick and timely data transfer when faced with high data volumes.

Shorter cycle times and faster processes in industry often require temporal synchronization of all participants within a system. The Precision Time Protocol (PTP Version 2, IEEE 1588) is supported by the new Ha-VIS mCon 3000 switches, and enables synchronization in the microsecond range. As a result, in many cases there is no need for proprietary automation solutions specially designed for real-time functioning. Extensive software features allow users to perfectly configure switches for each application scenario.

The new switches mean that, besides the Rapid Spanning Tree Protocol (RSTP), the Medium Redundancy Protocol (MRP) is also available when configuring redundant network topologies. Network redundancy is the ability to survive link failure without a communication breakdown. HARTING mCon 3000 switches ensure high data communication availability and thus reliable industrial infrastructure. ■

□ IN BRIEF

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Module-neutral infrastructure concept for the Smart Factory

The revolutionary-like changes brought about by Integrated Industry have an impact on the specific design of industrial production facilities. Plants, systems and facilities are increasingly built in modular form. Modular plant structures require a module-neutral infrastructure, i.e. one that is generic. In addition, the future infrastructure of Smart Factories will be scalable, highly available, and offer all necessary topologies.

□ Andreas Huhmann, Strategy Consultant Connectivity + Networks, HARTING Technology Group, Andreas.Huhmann@HARTING.com

WHY DO SMART FACTORIES REQUIRE MODULE-NEUTRAL INFRASTRUCTURES?

The reason can be deduced from the technological changes brought about by Integrated Industry. Industrial automation got under way with the fieldbus and without module-neutral infrastructure. The task of the fieldbus was originally limited to collecting I/O information in a decentralized manner. Here, individual I/O modules still failed to make comprehensive services available within the production process □ it was only about saving on wiring to the sensors and actuators.

In a radical departure, service in Integrated Industry production proceeds from the vantage point of the product to be produced □ which may include mechanical processing, printing or product packing. In a Smart Factory, these tasks are performed by production modules, which in turn use a communication system □ the fieldbus □ to create an internal structure. As such, the generic infrastructure of a modular plant is located outside the actual manufacturing

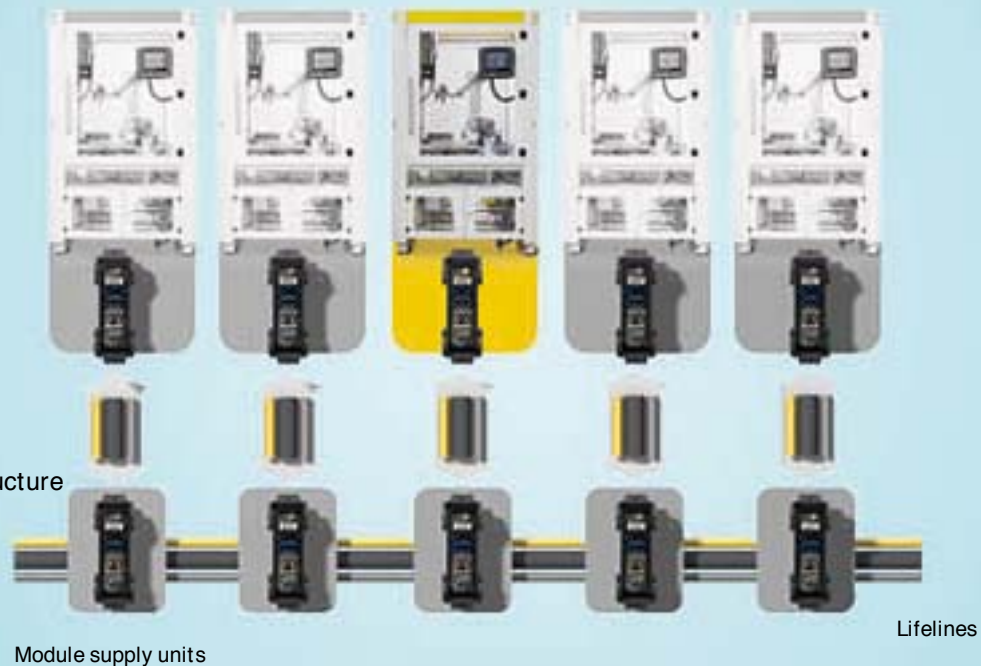
module, whereas the fieldbus is found primarily within the module. During the era of non-modular design, these limits were non-existent. The fieldbus was therefore universally employed.

In the era of Integrated Industry, the individual modules and the service they provide stand in the foreground. These modules can be supplied by different manufacturers. They are flexible and variable, meaning they can be deployed at different sites and in different production processes to produce different products. Connection to vital lifelines can be largely standardized. In order to make flexible integration possible, module-independent infrastructure is required.

Thus, this approach differs significantly from the decentralization approach of fieldbuses. However, synergies do exist. The use of convergent networks offers huge advantages for the integration of IT down to the field level, as the result is simple and fast data exchange between IT and automation □ i.e. Integrated Industry.

Modular plant structures

Module-neutral infrastructure



WHAT DOES MODULE-NEUTRAL INFRASTRUCTURE HAVE TO OFFER?

The infrastructure offers each module a communication link to all other modules in tandem with all applications, as well as supplying necessary media such as power, air and safety.

” In the era of Integrated Industry, the individual modules and the service they provide stand center stage.

The infrastructure is scalable and highly available, while also providing the necessary typologies. The infrastructure's functionality must be module-independent and possess standardized interfaces. Distributed line topology and the module independence of the infrastructure lead to active □module supply units□from which 1-2 modules can be supplied and/or which can be branched into other lines.

For each application-specific infrastructure configuration, HARTING offers the □Module Supply Units□the appropriate components from the Han-Modular□ system all the way to Ethernet network components. The HARTING smart Power Networks Unit comes with a 10-port Ethernet switch that features SFP slots and is also equipped with the appropriate interfaces and enables energy data acquisition along with central data storage and analysis. ■

The connector □ the foundation of Integrated Industry

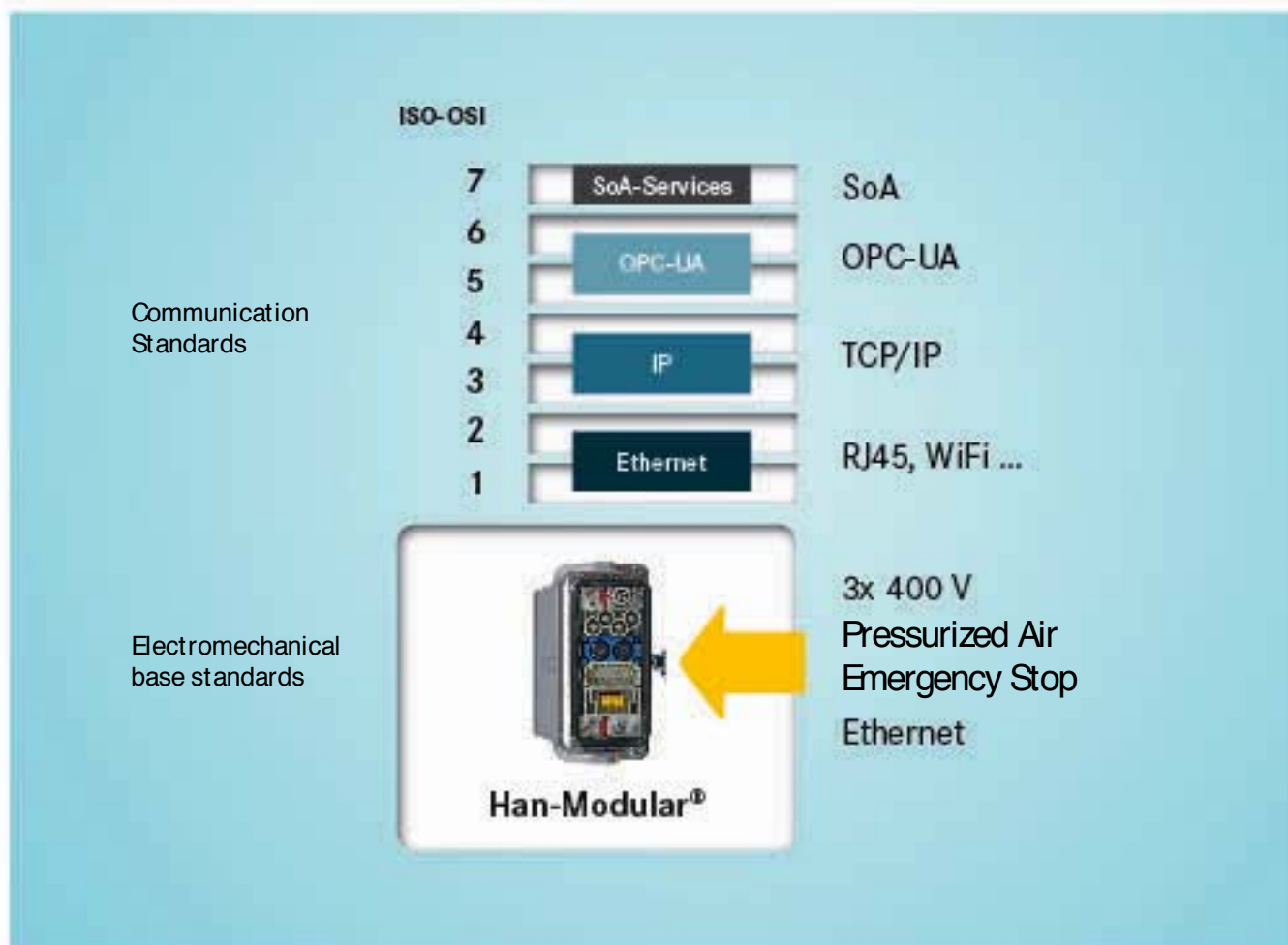
Smart Factory system modules require a simple connection to infrastructure. Han-Modular[®] means these connections are produced uniformly while still offering customization.

□ John Witt, Sales & Business Development Manager, HARTING Technology Group, John.Witt@HARTING.com

Integrated Industry is often viewed solely from the perspective of IT technologies and therefore seen as a pure software issue □ but it is no less dependent on the appropriate hardware components. In particular, these include the electro-mechanical interfaces that provide connection to the infrastructure, i.e. the connector foundation.

MODULAR CONNECTORS

A Smart Factory calls for the use of a standardized, modular connector. The connector must be suitable for all media present within a system module as well as the media to be supplied externally. Consequently, the connector functions as a power connection, a communication port (Gigabit Ethernet),





signal connections for various auxiliary connections, and for prepared compressed air for supplying handling units. A size 16 connector is used in the Smart Factory. The connector has four module slots.

The dimensioning of the connectors is based on a uniform standardization of the interfaces. Here, Han-Modular□ meets all the requirements of the system modules across the broad spectrum of modules available.

Besides the system module, in many cases the backbone connection must be implemented in pluggable manner as well. Given this, within the backbone the need will exist for the system-specific determination of the connectors since the dimensions of the backbone depend on the requirements of the overall system. Here, key criteria are total power consumption as well as spatial extension. The correct dimensioning of cables and contacts must be carried out for the power supply. With regard to communication, the distance involved will determine the selection of electrical and optical transmission.

Han-Modular□

The Han-Modular□ range offers a variety of combinations for transmission media. This allows the required connections to be dimensioned in space-saving and cost-conscious manner and permits the connectors to be adjusted to the respective installation situation. The product range includes over 40 different modules that can be combined for the transmission of power, signals, data and compressed air.

The modules can be used in a wide variety of HARTING connector housings with the help of special plastic and metal frames. Contacts and inserts for power up to 200 A / 1000 V and high voltage modules handling up to a maximum of 5000 V / 40 A are available. In addition, signal modules are available which



feature high packing density that extends up to a maximum of 25 contacts per single module (50 □ 250 V / 4 A □ 10 A).

” Han-Modular□: A Smart Factory calls for the use of a standardized, modular connector.

Moreover, the range extends from Ethernet Cat. 6 interfaces to modules for single and multi-mode fiber optic cable transmission. In addition, various bus technologies are implementable, e.g. PROFIBUS. Even pneumatic contacts can be integrated which transmit prepared compressed air up to 8 bar. Numerous other application possibilities round out the product lineup.

PLUGGABLE FIBER OPTIC CONTACTS

The LC module for pluggable fiber optic contacts ranks among the most recent additions. Fiber optic cable offers enormous advantages such as the transmission of sensitive industrial data. It effectively shields the transmission from sources of external interference such as electromagnetic radiation. It also enables galvanic transmitter-receiver separation and prevents cross-talk. ■

From sensors to the Cloud and back

It is common practice in modern production facilities to record sensor data in order to control production processes and monitor system units. But what would happen if, within a fraction of a second, this sensor data could be stored and subsequently become available again for further analysis □e.g. to predict failures or unplanned outages? Precisely this scenario forms part of Integrated Industry.

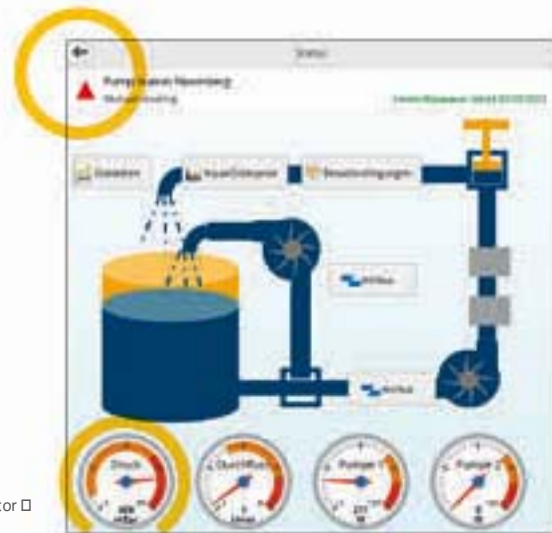
□Marco Lampe, Head of Softwaredeveloping Auto-ID, HARTING Technology Group, Marco.Lampe@HARTING.com



Predictive maintenance

The effects would be far-reaching, and the organizational and economic benefits are striking. Consequently, HARTING Technology Group has developed a □Predictive Maintenance□demonstrator together with SAP based on the SAP HANA database Cloud technology. The initial considerations for the development of a production-oriented demonstrator for Industry 4.0 were formulated in June 2013. This was complemented by predictive maintenance considerations slated to be based on an SAP HANA database.

Working on the basis of these two specifications, a complete □water works□featuring pumps, water circulation, sensors and an embedded micro-computer with management control software designed by HARTING was developed within a period of six weeks. The aim of this demonstrator is to test the robustness of a concept under essentially real conditions and in which errors in processes or components can be detected early on in order to order



Representation of the state of the demonstrator □ a system failure has been diagnosed

spare parts in advance, minutely plan replacement work, and be able to make appropriate remediation measures available to technical service personnel.

Inside the □waterworks□water is continuously pumped out of a container and back into it. Sensor data from the pressure and flowmeter are queried at extremely short intervals by the embedded micro-computer and then relayed to the SAP HANA

cycle that shows signs of failure can be disabled and switched over to a second one that is also integrated into the demonstrator. This permits the malfunction to temporarily be bypassed.

VISUALIZATION DURING SERVICE AND MAINTENANCE

To resolve the error, in addition to the information on the faulty component, the technician also has access to a detailed

pass maintenance and repair. The RFID transponders permit components to be uniquely identified, and the history of every component can be documented. As a result, maintenance intervals can be precisely planned down to the component level. ■

” Maintenance intervals can be precisely planned down to the component level.

database. In the event that a change in sensor data is registered with respect to the circulation, an algorithm stored in the SAP HANA database can mathematically and precisely determine the instant of the potential failure.

A graphical interface is employed to display status messages that can be used to analyze errors. In addition, a pumping

3D drawing based on current CAD diagrams, which can assist in performing replacement work.

Further development of the demonstrator is planned: with the help of HARTING RFID transponders attached to each component, in tandem with Ha-VIS Middleware and the Ha-VIS Application-Suite, the scenario could be extended to encom-

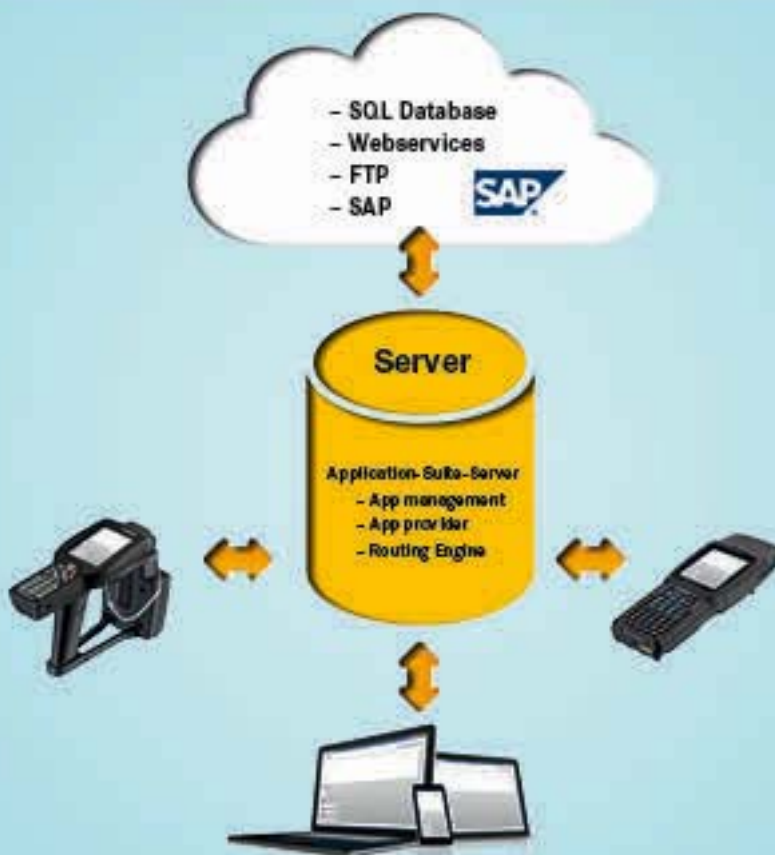
ON BRIEF

Conversion of unplanned maintenance into planned maintenance

Expertise in app solutions: the Ha-VIS Application-Suite

How does the right app find its way onto a service technician's mobile reader? Who makes sure that the required process steps are mapped and depicted? How should such requirements be efficiently programmed and maintained from a central location? HARTING's solution goes by the name of the Ha-VIS Application-Suite.

□ Olaf Wilmsmeier, Product Manager Software, HARTING Technology Group, Olaf.Wilmsmeier@HARTING.com



Principal architecture of the Ha-VIS Application-Suite

A work shift gets under way in the maintenance and service department of a factory. We accompany a technician who performs maintenance on equipment within the company. This activity must be thoroughly documented and centrally recorded in an in-house database system.

But the technician doesn't grab a folder full of checklists from the shelf that details the work he needs to complete today. Instead, he picks up a mobile RFID reader with integrated touch panel. After logging on to the device, his work can begin.

RFID IDENTIFICATION

Arriving at the first maintenance location, the machine scheduled for servicing is uniquely identified by UHF RFID with the aid of the mobile RFID reader. The maintenance checklist is automatically shown to the technician on the display, including the correct machine/object identification. Maintenance work is digitally documented directly on the mobile recording device. Thanks to



RFID identification, any mix-up with the wrong checklist or an incorrect maintenance object is ruled out.

can be updated and maintained there. The predominantly mobile client devices receive the necessary information and

an app that is created can run on different end devices supported by HARTING.

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The Ha-VIS Application-Suite extends the HARTING Auto-ID product portfolio. Naturally, the Ha-VIS Application-Suite is compatible with the powerful Ha-VIS Middleware, which is compliant with the EPCglobal ALE 1.1 standard. ■

PRACTICAL QUESTIONS

But who ensures that the recorded data are stored in the correct database? How complex is the maintenance of such a system? What happens if different recording devices are used? Can an application potentially be adapted to new work processes in flexible manner? Does the mobile recording device need to be permanently connected to the server?

HARTING Ha-VIS Application-Suite takes over and organizes these tasks. The client-server based architecture of this solution platform makes it possible to create hardware-independent and operating system-independent applications (apps). In concrete terms, this means that an app runs on a central server and

interfaces from this central location. But unlike a conventional client-server based mobile app, more than just a website is displayed on the client. Extensive job steps such as maintenance work can be performed offline. The mobile device loads necessary data once into memory and can then continue to work offline. In addition, in contrast to pure Web-based applications, the hardware of the mobile end device can be accessed □ for example, that of the RFID reading unit.

To accomplish this, HARTING makes device-specific connections available. As a result, during the actual app creation the device-dependent properties can be ignored. The advantage of this technique:

IN BRIEF

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Railway applications:

HARTING connectors meet new fire protection standard



The standard EN 45545-2 "Railway applications - Fire protection on railway vehicles" means that, for the first time, a mandatory European requirement for the fire behavior of components and materials used in rail vehicles is now in place. The standard also takes into account installation situations.

□ Christoph Dossow, Market Manager, HARTING Technology Group, Christoph.Dossow@HARTING.com

□ Malte Hofmann, Industry Segment Manager, HARTING Technology Group, Malte.Hofmann@HARTING.com

The new standard specifies test methods and limit values and establishes requirement sets (R1 to R26) for components. Hazard levels HL1 to HL3 indicate the testing severity.

Small electrical components like electronic connectors must demonstrate a material certification rating of V0 which represents very low flammability. No obligation to demonstrate certification exists for flammable materials under 10g in weight unless they are installed adjacent to components for which no certification exists. In such cases, the requirements depend on grouping rules. HARTING plastic housings and inserts in connectors fulfill V0 requirements.

Depending on the installation situation, the same requirements may be in effect for heavy connectors as for the switch cabinet on which they are mounted. If the cabinet wall acts as a fire resistance zone, depending on its size the cabinet must exhibit a fire resistance of E10 or E15. The number represents the minutes that a connector must withstand as a physical barrier in case of fire. If the requirements are fulfilled, the components in the interior are exempt from the obligation to demonstrate certification.

HARTING housings in the Han[□] B, Han[□] M and Han[□] HPR series meet the requirements of E15. Most HARTING housings even resist fire for more than 30 minutes. Additionally, their surface coating meets the requirements for flame spread.

For the materials in connectors, R22/R23 are the maximum applicable requirement sets. The sets prescribe the parameters, procedures and limits for tests. Specifically, R22/R23 mandate tests and limit values for oxygen content, smoke density and toxicity. The polycarbonate used by HARTING in its connectors and special electronics products conforms to the limit values. ■

IN BRIEF

t EN 45545-2

t Uniform pan-European requirements

t HARTING products meet the strictest requirements



Model Class

Operating Class	N: Standard vehicles	A: Vehicles for automatic drive operation which do not have trained emergency personnel on board	D: Double decker vehicles	S: Sleeper car and couchette car
1	HL1	HL1	HL1	HL2
2	HL2	HL2	HL2	HL2
3	HL2	HL2	HL2	HL3
4	HL3	HL3	HL3	HL3

Test Method		Parameter	Unit	Internal	External
Oxygen index	EN ISO 4589-2	OI	%	F22	F23
Smoke density	EN ISO 5659-2	Ds max.	No unit	F22	F23
Smoke toxicity	NFX 70-100-1/-2	CITNLP	No unit	F22	F23

Test Method	Parameter	Unit	Threshold (R22)			HARTING (polycarbonate)
Oxygen index	OI	%	HL1: 28	HL2: 28	HL3: 32	F22, F24 = HL3
Smoke density	Ds max.	No unit	HL1: 600	HL2: 300	HL3: 150	F22 = HL3
Smoke toxicity	CITNLP	No unit	HL1: 1,2	HL2: 0,9	HL3: 0,75	F22 = HL3

Han[□] HMC[□]

Higher mating cycles for modular industrial connectors

The modular construction of industrial facilities is developing into the central solution in Industry 4.0. This also increases the demands made on heavy-duty connectors. With the Han[□] HMC, HARTING is introducing a connector family designed for over 10,000 mating cycles.

□Gero Degner, Product Manager, HARTING Technology Group, Gero.Degner@HARTING.com

□Industry 4.0□is the central concept with which classical industry is evolving at a considerable speed. A key point in this development is the strong individualization of products, which nevertheless requires highly flexible mass production.

This means that, in addition to more intelligent control and automation technology, highly flexible interfaces are being required in more and more modularly constructed production facilities. These interfaces are being faced with greater demands than hitherto has been the case due to more frequent tool and system change intervals. In particular, they need to accommodate increased mating cycles.

Based on these aspects, the HARTING Technology Group has developed the Han[□] HMC series as heavy-duty industrial connectors. While standard industrial connectors are designed to

handle in excess of 500 mating cycles, the Han[□] HMC series tolerates more than 10,000 mating cycles with no performance degradation. Even after such high loads, the requirements of DIN EN 61984 continue to be met.

TECHNICAL BASIS

The Han[□] B series forms the technical basis for this new connector, which has proven itself millions of times over and has been firmly established on the market for decades. Focused development efforts have seen the Han[□] B series successfully adapted to these new requirements. In this context, new solutions were employed. By way of example, entirely newly developed high-performance contact springs replace conventional earthing contacts. Likewise, special measures permitted the locking system to be adapted to the somewhat increased number of locking operations.

” The Han[□] HMC series extends the product portfolio for industrial applications that require more than 10,000 mating cycles for the connectors they deploy.



The contacts are the heart of a connector, which must reliably transmit signals, data, and power over its entire lifespan. The Han D[□] and Han E[□] contacts employed in the Han[□] HMC series are coated with a special gold layer, ensuring optimum electrical properties in all stages of the connector's lifespan.

For customers, however, nothing changes in terms of mounting dimensions and the assembly tools to be used. This makes a transition to the new series equally convenient and unproblematic, since the connectors can be used in the customary way. ■

IN BRIEF

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High Speed 100 Gigabit technology



Today, industrial networks typically operate based on Fast Ethernet with 100 Mbit/s and are moving towards Gigabit Ethernet in order to be able to meet the ever-growing demands made on network technology. The HARTING Technology Group is intensively driving this development process forward - in the standards committees as well as in its laboratories and product development efforts.

□ Rainer Schmidt, Business Development Manager, HARTING Technology Group, Rainer.Schmidt@HARTING.com

No other technology influences the development of society and the economy as intensely as ICT (information and communication technology). The expansion of communication networks is continuously and permanently changing everyday communication behavior. At the same time, ICT itself is becoming a driver for new applications in public administration, transport, industry and science.

In industrial applications, this is relevant with regard to two considerations:

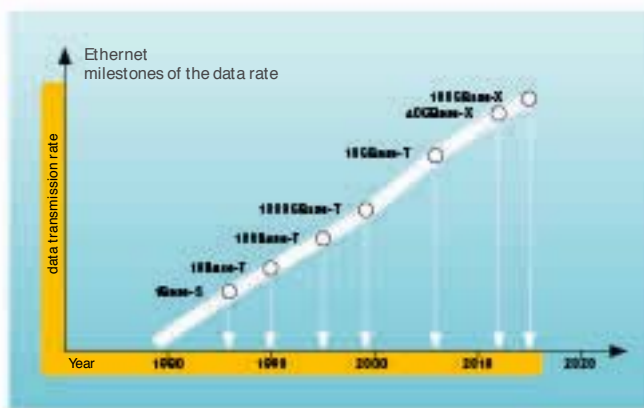
For one, possibilities arise for the modernization and rationalization of production processes, which enhances efficiency and further reduces cost burdens. Example: automation using the latest generation of PROFINET.

Secondly, possibilities open up to equip products themselves with additional features, which results in additional competitive advantages. Example: Ethernet-based passenger information and entertainment systems in long-haul trains.

Consequently, from the perspective of the HARTING Technology Group, information technology - and especially Ethernet and Ethernet cabling - represents a key technology.

In order to advance the development of Ethernet, the IT industry has organized itself in the form of a global body - the IEEE (Institute of Electrical and Electronics Engineers).

New and more powerful device technology (switches, routers, controllers) calls for adequate cabling. As a consequence of this dependence, cabling for this device technology needs to take the lead technologically, i.e. it must continually be capable of handling higher transmission capacities and speeds than those used in the current state of development.



Development of bandwidth in data networks



For this reason, **HARTING** is investing heavily in the development of new cabling technologies, which in turn flows into new products such as termination technology, connectors, cables and cords. However, the technical foundation and preconditions must be created for this to happen. Currently, efforts are focusing on the search for solutions for the transmission of 100 Gigabit Ethernet over a channel with symmetrical copper cables and connectors.

RESEARCH PROJECT

In order to meet the challenges, **HARTING** has initiated a research project funded by the Federal Ministry for Research and Technology. The project group comprises experts from industry and the University of Reutlingen. The project has been tasked with devising a 100 Gigabit channel based on simulation models, and subsequently demonstrating its functionality. The parameters thereby obtained will then be examined in **HARTING** laboratories. After consolidating all the results, the IEEE will



Latest generation balun-less measurement technique in **HARTING** labs

” **HARTING** Technology Group is investing heavily in the development of new cabling technologies, which in turn flows into new products such as termination technology, connectors, cables and cords.

receive a recommendation for the development of the new 100 Gigabit Ethernet standards for transmission over a defined 100 GBase-T cabling system.

This initiative ensures that **HARTING** customers enjoy early and rapid access to the latest technologies and products related to Ethernet cabling. ■

IN BRIEF

Competitive advantages through new technologies such as 100 Gigabit Ethernet

t Cost savings in the production process thanks to efficient automation

t Faster cycle times in production based on a unified data network from the data center to plants and facilities



Further Information

Cable assemblies?



Overmolded cable connections are an attractive option in industrial applications due to their usability in harsh environments. In the context of increased flexibility, interest is becoming stronger in retaining the advantages of overmolded solutions while still responding to individual requirements. Flexibility and safe use in harsh environments are not contradictory, as demonstrated by HARTING Technology Group taking its M12 cable assemblies as an example.

□ Dirk Peter Post, Product Manager, HARTING Technology Group, Dirk-Peter.Post@HARTING.com

In the industrial environment, so-called overmolded solutions for cable assemblies are deployed and accepted worldwide. This proven technology is attractive, as long as the same types of cables are processed in large quantities. Their cost is lower than that of bolted solutions, and they also demonstrate better performance in harsh environments.

The markets, however, tend to use different types of cables. For overmolding, this means repeatedly adjusting the tool employed. In addition, many customers rely on their own cable specifications and therefore on their own cables. Last but not least, widely differing types of cables are deployed in similar application environments during set-up. This results in an increasing emphasis on short-term project needs.

HARTING has risen to this challenge and developed a customer-oriented, flexible, safe and widely deployable connection technology that approximates the benefits of overmolded solutions. Here, efficiency and flexibility are at the forefront.

THE HARTING SOLUTION

The solution chosen by HARTING makes use of the already available elements from the M12 Slim Design. The big difference is that tightening is no longer required. Instead, a cap is pressed on □ and it's ready. In a word: just Press and Go!

This HARTING innovation offers a non-detachable, custom cable assembly that can be deployed quickly and economically. The mechanical and electrical properties are equivalent to the overmolded solution. Another advantage is that all cables can be used in a diameter range of 4.5 to 8.8mm, and tool adjustments

” HARTING has risen to this challenge and developed a customer-oriented, flexible, safe and widely deployable connection technology that approximates the benefits of overmolded solutions. Here, efficiency and flexibility are at the forefront.

are not required. The technology is distinguished by its quick time-to-market and flexibility with regard to cables customized to customer requirements.

The first products featuring the M12 technology will be on display at the Hannover Messe 2014 trade fair. ■

IN BRIEF

- t □ Can be safely used even in harsh environments
- t □ Cost efficient production
- t □ Fast, customized availability

Selectively loaded har-flex[□] connectors offer even more flexibility in device design

There are trends that begin in machine and plant construction and then make their way from control and drive technology all the way to a device's internal interface. In a word, modularization, miniaturization, flexibility. HARTING now fulfills these requirements to an even greater extent thanks to the expansion of the har-flex[□] product family to include selectively loaded variants.

□Joachim Finke, Product Manager har-flex[□], HARTING Technology Group, Joachim.Finke@HARTING.com

The trend in machinery is towards small, independent modules capable of performing individual work steps. The modules feature decentralized control and drive technology and can be easily combined with one other. This increases the flexibility available to mechanical engineers.

The demands on automation technology are growing in step with this. No matter whether control or drive units are involved □the demands are for ever smaller, more powerful units. Tailored functionality for each machine module is expected.

More and more, this is being implemented via modularly constructed automation systems. Whether Control units, pneumatic valve terminals or frequency converters, many new and innovative device generations can be flexibly extended by pluggable function assemblies and thus be adapted individually to the requirements of applications.

This is only enabled by device-internal interfaces that, in addition to miniaturization, provide targeted support for modularization and flexibility in the device design.

This is precisely where the HARTING har-flex[□] product family of electronics connectors enters the picture.

INDUSTRY APPROPRIATE MINIATURIZATION

The small 1.27 mm grid and an extremely compact design permit miniaturized device design. Despite its dimensions, the connector delivers what HARTING products promise: High reliability also under harsh industrial conditions. Consequently, the side-mounted SMT hold-downs relieve the signal contacts from mechanical stress. This ensures the reliability of the overall system - even at high shock and vibration loads.

” The small 1.27 mm grid and an extremely compact design permit miniaturized device design.

NEW PATHS TO MODULARIZATION

This robustness, coupled with mating cycles in excess of 500 cycles, opens up entirely new paths in the modularization of industrial equipment. As a result, har-flex[□] connectors are already being used in highly innovative device generations as an interface for additional function assemblies that can be plugged and unplugged in the field by users. In essence, they act as an external interface.



BOOSTING FLEXIBILITY EVEN MORE

The availability of straight and angled male and female connectors, as well as IDC cable assemblies in all the even number position pin counts from 6-100, means there is no limit to flexibility in device design.

But **HARTING** is now driving flexibility considerations even further ahead.

Effective immediately, all straight and angled male and female connectors are available as selectively loaded versions. With the option of partial loading, customers can increase clearance

and creepage distances and thus flexibly adapt to their system requirements. Where two or more connector pairs previously needed to be used, electrical isolation is now possible within an interface. This reduces costs, increases flexibility in equipment design, simplifies and shortens the SMD loading process and also saves valuable board space.

And partial loading is not subject to limits □ **HARTING** can already implement any type of partial loading upon customer request. ■



IN BRIEF

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t **SVTUPNJ FETFMIDJWVMBECHOPTTJOM**

Robustness + harsh environment = M12



The M12 connector, which has proven itself in industrial applications, is also a top performer in outdoor applications □ thanks to use of specific materials.

□ Dirk Peter Post, Product Manager, HARTING Technology Group, Dirk-Peter.Post@HARTING.com

Industrial environments would be unimaginable without the classic M12 connector. Data, sensor and power applications □ the M12 masters them all. The connector's manufacture relies on well-known materials and production processes: die-casting, injection molding and the rotational technique are all part of the job.

Increased flexibility and the decentralization of facilities is entailing the rising necessity to deploy connectors in harsh environments as well as outdoors. Today, connectors are expected to feature adequate encapsulation against salt spray and mist, which in turn requires more resistant materials.

DURABLE MATERIALS

For this reason, the HARTING Technology Group tested the use of especially durable materials. Stainless steel in V2A quality is not sufficiently resistant, and the appropriate higher grade material is referred to as V4A.

Processing V4A, however, is problematic: series production readiness and reliable machining are not guaranteed using conventional machines and tools. The elements of material and tool must be coordinated with one another. The material itself must meet the most rigorous standards, which is not the case for some of the materials on offer. While they are easy enough

” HARTING M12 crimp connectors in V4A are protected from extreme salt spray and mist.

to process, they fail to demonstrate adequate resistance in the field.

On the manufacturing side, the HARTING Technology Group has succeeded in tackling V4A grade stainless steel. As a result, this means that M12 crimp connectors in V4A are protected from extreme salt spray and mist. HARTING has proven this in tests that conform to IEC 60068-2-52 severity 4. Consequently, this has demonstrated that the M12 connectors are robust and suitable for deployment in the harshest of environments. ■

□ IN BRIEF

Deployable even under extreme environmental conditions, such as outdoor scenarios

HanOnBoard[□]

Relaying data, signals and power is an essential prerequisite of industrial production. Decentralized distribution boxes equipped with Han[□] connectors perform this task. HARTING Technology Group now offers an alternative to the manual wiring of Han[□] connectors in the box.



□ Dr. Joerg Wissdorf, Managing Director HARTING Integrated Solutions, HARTING Technology Group, Joerg.Wissdorf@HARTING.com

The standard solution for distribution boxes is contacting individual connectors via manual wiring (Figure 1). Manual wiring, however, is associated with significant drawbacks: the substantial wiring effort needed in production and the high susceptibility to harsh outdoor conditions such as shock and vibration are considerable. Troubleshooting the system is not efficient, while there is also the risk of secondary faults in service.

The aim of HARTING's development effort was a solution that permitted the continued use of proven Han[□] products while making it possible to avoid the disadvantages of manual wiring.

HARTING's solution is called HanOnBoard[□]. In this solution approach, Han[□] connectors are connected with a PCB (Printed

Circuit Board) using an adapter via which data, signals and power are then quickly and reliably distributed (Figure 2).

The solution entirely dispenses with time-consuming and potentially fault-prone wiring and is based on an industrially manufactured combination composed of the tried and tested Han[□] components, HARTING PCB adapters and the HARTING PCB instead.

HanOnBoard[□] is compact and weight-saving, while susceptibility to shock and vibration is reduced to a minimum. Tedious troubleshooting is replaced by modern, quick module replacement. Secondary service faults are avoided.

In additional, special technical processes ensure that sufficient flexibility remains for on-site adjustments also when PCBs are used.

All HanOnBoard[□] components come from HARTING. Since the design of the overall solution also takes place at HARTING, only a single contact partner is needed, which significantly reduces the time required to produce a solution. ■



Conventional wiring in a box



HanOnBoard[□] solution □ space-saving, robust, simple to manufacture and service

For questions, please feel free to call us at
+49 (571) 8896-501.

One switch, one cable for power and data

The seamless integration of new technologies in industrial production systems requires intelligent and reliable supply of data and power. Among other features, HARTING Technology Group's newly developed eCon switches offer high-performance Power over Ethernet for powering industrial equipment.

□ Oliver Opl, Product Manager, HARTING Technology Group, Oliver.Opl@HARTING.com

Visual systems are becoming increasingly integrated into industrial manufacturing processes, as monitoring and control via camera systems enables process flexibility and ensures product quality. Besides the need for stable and efficient data communication, the systems require a reliable power supply. HARTING Technology Group's newly developed Ha-VIS eCon switches provide a completely integrated solution for data and power transmission in a single device, along with a simultaneous reduction of the cabling effort and expense. The enormous number of more than 200 different switch variants offers the opportunity to specially select a specified switch for each individual application. Two different, compact and space-saving housing designs guarantee the best possible use of the space available inside the switch cabinet.

Their compact form means that the switches in the Ha-VIS eCon 3000 family permit very high packing density on the DIN rail in the switch cabinet, while the Ha-VIS eCon 2000 switches are characterized by their exceptionally flat design. As a result, the eCon series offers features that enable efficient system design.

POWER OVER ETHERNET AS THE SUPPLY SYSTEM

Power over Ethernet (PoE) allows the transmission of data and power by way of a single cable, i.e. over the standard Ethernet data cable. Switches that can supply end devices with energy in this way are referred to as PSE (Power Sourcing Equipment), with the end devices supplied with power designated as PD (Powered Device). Under the current standard, PoE provides a minimum of 25.5 watts to end devices.

” Besides several new features, HARTING Technology Group's new eCon switches also offer high-performance Power over Ethernet for powering industrial equipment.

Next generation Ha-VIS eCon switches support the current standard and can simultaneously make full PoE+ power available to up to four ports. In total, then, each switch provides over 100 Watts of power to any type of end device. As a result, in addition to IP phones and powerful Wi-Fi access points, high resolution camera systems can also be supplied with power.



RAPID RETROFITTING

Existing applications can be retrofitted extremely easily since there is no need to lay separate cables for the power supply. This reduces both the cost and effort associated with installation. On the other hand, existing installations can typically continue to be used and retrofitted. Only the PoE+ switches need be added. Beyond the PoE+ standard, the variants in the Ha-VIS eCon portfolios also offer the option of an integrated DC/DC converter. Here, the switches can be supplied with 24V DC (direct current) instead of the normally required 54V DC. While no special RJ45 plugs are required for power transmission, users should consider their suitability for industrial applications when selecting cabling components or systems, in particular their suitability for PoE. As a specialist in industrial data network technology, HARTING has designed RJ45 components, system cords and cables specifically for parameters such as current-carrying capacity, temperature range and vibration resistance. These components are specifically tested for use with PoE and PoE+, guaranteeing users long service life and high operating reliability. ■

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Dynamic RFID race





HARTING and the Ignition Racing Team of the Osnabrück University of Applied Sciences' conduct an RFID High Speed Test at Porta Westfalica airport.

How fast may a vehicle go and still be detected with maximum reliability using RFID? This question is highly relevant in real-world practice, as demonstrated by areas of application such as rail transport. The HARTING RFID team joined up with Osnabrück University of Applied Sciences' Ignition Racing team at the Porta Westfalica airport to investigate this question. The result: even at 200 kmh, a racecar was quickly and reliably identified by an RFID transponder □ with the potential for further increases.

□ Dr. Jan Regtmeier, Product Manager RFID Reader, HARTING Technology Group, Jan.Regtmeier@HARTING.com

HARTING RFID systems are extremely fast. But how fast is fast? The answer to this question is highly relevant for real-world applications, since modern warehouse systems move objects at speeds exceeding 50 kmh and today's freight trains can reach up to 160 kmh.

Reliable detection and error-free identification of goods and vehicles is essential for process safety at these high speeds □ and high process reliability is the only way to ensure time and cost savings. Data must be collected quickly and safely in order, for example, to specifically direct goods onto a different conveyor belt or activate signals in railway environments.

The RFID team at HARTING Technology Group investigated the question of □ how fast is fast □ in a joint effort with the Ignition Racing Team (IRT) of the Osnabrück University of Applied Sciences. As part of the Formula Student project, the IRT develops and builds their own racing cars. The performance metrics are spectacular: up to 1400 Nm of torque accelerate the race cars from zero to 120 kmh on a 75 m track □ a normal car requires a multiple of this.

The experimental setup for the HARTING RFID test was exceptionally simple: each

race car was equipped with an RFID transponder. An RFID reader and antenna was built alongside the track. And, in order to obtain realistic conditions, only standard products were used. In addition, the results were evaluated in real time in Ha-VIS Middleware and written to a file □ in exactly the same setup as would be implemented in an actual customer application.

” The racecar was detected nine times at 200 kmh.

TEST RUN

As a warmup exercise, an electric racecar drove past the RFID antennas at 80 kmh. All systems functioned successfully and the car was reliably detected. The second race: an electric racing car at 120 kmh □ no problem here either, as the vehicle was reliably detected. However, the electric racecars top out at a maximum speed of 120 kmh since the Formula Student project only uses a narrow track or a 75 m sprint. Consequently, in the third round a race car with a combustion engine was used, reaching a top speed of 160 kmh (which is extremely fast for the Formula Student racecars).

Still, the test run failed to push the RFID technology to its limits. Therefore, a roadster of a famous German manufacturer was driven in one last run. The result was startling: even at 200 kmh, the vehicle was quickly and reliably detected! In technical terms: a unique number (Electronic Product Code or □ EPC □) with a length of 96 bits was successfully read nine times. This proves the functionality of RFID technology at top speeds as well, meaning the HARTING RFID team will now head out in search of new challenges. ■

□ IN BRIEF

Keep an eye out for our RFID high-speed video on our HARTING Youtube channel!



How did this issue perform?

Dear Readers,

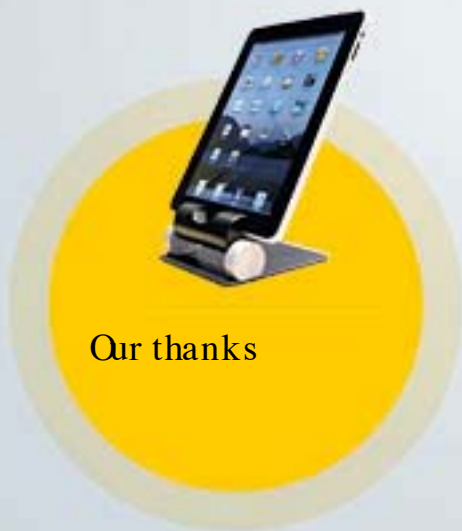
Just like Robert HARTING – European Champion, World Champion and Olympic Champion in discus throwing – we too strive for consistent further development, and aspire to delivering absolute peak performance.

Consequently, we're interested in your opinion! Because your feedback helps us continually make tec.News just a little bit better. And of course, we're particularly pleased to hear any praise you might have. Our goal is to keep you informed of new developments and applications at HARTING – in the best possible way. We are only asking you for a few minutes of your time to take part in our online survey.

As our way of saying thank you, an iPad will be raffled off to one lucky survey participant. You can take part in the online survey until July 31, 2014.

Thank you so much for your support.

Your tec.News editorial team



This link will take you directly to the reader survey:

www.HARTING.com/tecNews-survey



HARTING Trade Show Calendar

07.04. □ 11.04.2014	Germany, Hanover, Hannover Messe
12.04. □ 15.04.2014	Korea, Seoul, Automation World 2014
22.04. □ 25.04.2014	Ukraine, Kiev, Elcom Ukraine
23.04. □ 26.04.2014	China, Shanghai, Chinaplas 2014
06.05. □ 08.05.2014	Austria, Vienna, Smart Automation
14.05. □ 15.05.2014	Belgium, Brussel, Automation & Engineering
20.05. □ 24.05.2014	Malaysia, Kuala Lumpur, AutoMex 2014
20.05. □ 24.05.2014	Brazil, São Paulo, Feira Mecânica
20.05. □ 22.05.2014	Italy, Parma, SPS/IPC/DRIVES
03.06. □ 05.06.2014	Germany, Nuremberg, SENSOR+TEST
16.06. □ 20.06.2014	France, Paris, Eurosatory
17.06. □ 19.06.2014	Great Britain, London, IFSEC Exhibition
17.06. □ 20.06.2014	Russia, St. Petersburg, Energetika&Elektrotekhnika-2014
10.08. □ 14.08.2014	USA, Orlando, FL, RFID Journal Live!
02.09. □ 04.09.2014	Switzerland, Bern, Sindex
16.09. □ 20.09.2014	Netherlands, Amsterdam, ECOC
23.09. □ 26.09.2014	Germany, Hamburg, Wind Energy Hamburg
23.09. □ 26.09.2014	Germany, Berlin, InnoTrans
29.09. □ 03.10.2014	Czech Republic, Brno, MSV 2014 Brno
07.10. □ 09.10.2014	Sweden, Gothenburg, Scanautomatic
11.11. □ 14.11.2014	Germany, Munich, electronica

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Vice Chief Editor: Dr. F. Brode, A. Huhmann, Dr. S. Middelkamp

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AE - United Arab Emirates
HARTING Middle East FZ-LLC
Knowledge Village
Block 2A, Office F72
Dubai, United Arab Emirates
E-Mail: uae@HARTING.com, www.HARTING.ae

AT - Austria
HARTING Ges. m. b. H.
DeutschstraÙe 19, A-1230 Wien
E-Mail: at@HARTING.com, www.HARTING.at

AU - Australia
HARTING Pty Ltd
Suite 11 / 2 Enterprise Drive
Bundoora 3083, AUS-Victoria
E-Mail: au@HARTING.com, www.HARTING.com.au

BE - Belgium
HARTING N.V./S.A.
Z.3 Doornveld 23, B-1731 Zellik
E-Mail: be@HARTING.com, www.HARTING.be

BR - Brazil
HARTING Ltda.
Rua Major Paladino 128 □Pr□lio 11
CEP 05307-000 □SP Paulo □SP □Brasil
E-Mail: br@HARTING.com, www.HARTING.com.br

CA - Canada
HARTING Canada Inc.
8455 Trans-Canada Hwy., Suite 202
St. Laurent, QC, H4S1Z1, Canada
E-Mail: info.ca@HARTING.com, www.HARTING.ca

CH - Switzerland
HARTING AG
Industriestrasse 26, CH-8604 Volketswil
E-Mail: ch@HARTING.com, www.HARTING.ch

CH - Switzerland
HARTING AG Mitronics
Leugenstrasse 10, CH-2500 Biel 6
E-Mail: mit@HARTING.com
www.HARTING-mitronics.ch

CN - China
HARTING (Zhuhai) Sales Limited
Room 3501 - 3503
No. 1, Hong Qiao Road, Grand Gateway I
Xu Hui District, Shanghai 200030, China
E-Mail: cn@HARTING.com, www.HARTING.com.cn

CZ - Czech Republic
HARTING s.r.o.
MíChsk□2, CZ-160 00 Praha 6
E-Mail: cz@HARTING.com, www.HARTING.cz

DE - Germany
HARTING Deutschland GmbH & Co. KG
P.O. Box 2451, D-32381 Minden
Simeonscarri 1, D-32427 Minden
E-Mail: de@HARTING.com,
www.HARTING.de

DK - Denmark
HARTING ApS
Hjulmagervej 4a, DK □7100 Vejle
E-Mail: dk@HARTING.com, www.HARTING.dk

EE - Eastern-Europe
HARTING Eastern Europe GmbH
Bamberger StraÙe 7, D-01187 Dresden
E-Mail: Eastern.Europe@HARTING.com
www.HARTING.com

ES - Spain
HARTING Iberia S.A.
Avda. Josep Tarradellas 20-30 4o 6a, E-08029 Barcelona
E-Mail: es@HARTING.com, www.HARTING.es

FI - Finland
HARTING Oy
Teknobulevardi 3-5, FI-01530 Vantaa
E-Mail: fi@HARTING.com, www.HARTING.fi

FR - France
HARTING France
181 avenue des Nations, Paris Nord 2
BP 66058 Tremblay en France
F-95972 Roissy Charles de Gaulle □lex
E-Mail: fr@HARTING.com, www.HARTING.fr

GB - Great Britain
HARTING Ltd.
Caswell Road, Brackmills Industrial Estate
GF-Northampton, NN4 7PW
E-Mail: gb@HARTING.com, www.HARTING.co.uk

HK - Hong Kong
HARTING (HK) Limited
Regional Office Asia Pacific
3512 Metroplaza Tower 1, 223 Hing Fong Road
Kwai Fong, N. T., Hong Kong
E-Mail: ap@HARTING.com, www.HARTING.com.hk

HU - Hungary
HARTING Magyarorszá Kft.
Feh□v□ri □ 89.95, H-1119 Budapest
E-Mail: hu@HARTING.com, www.HARTING.hu

IN - India
HARTING India Pvt Ltd
7th Floor (West Wing), Central Square II
Unit No.B-19 Part, B 20&21, TVK Industrial Estate
Guindy, Chennai □600032
E-Mail: in@HARTING.com, www.HARTING.co.in

IT - Italy
HARTING SpA
Via dell'Industria 7, I-20090 Vimodrone (Milano)
E-Mail: it@HARTING.com, www.HARTING.it

JP - Japan
HARTING K. K.
Yusen Shin-Yokohama 1 Chome Bldg., 2F
1-7-9, Shin-Yokohama, Kohoku,
Yokohama 222-0033 Japan
E-Mail: jp@HARTING.com, www.HARTING.co.jp

KR - Korea
HARTING Korea Limited
#308 Yatap Leaders Building, 342-1 Yatap-dong
Bundang-gu, Sungnam-City, Kyunggi-do
463-828 Republic of Korea
PE-Mail: kr@HARTING.com, www.HARTING.co.kr

MY - Malaysia (Office)
HARTING Singapore Pte Ltd
Malaysia Branch, 11-02 Menara Amcorp, Jln. Persiaran
Barat, 46200 PJ, Sel. D. E., Malaysia
E-Mail: sg@HARTING.com, www.HARTING.com

NL - Netherlands
HARTING B.V.
Larenweg 44, NL-5234 KA's-Hertogenbosch
Postbus 3526, NL-5203 DM's-Hertogenbosch
E-Mail: nl@HARTING.com, www.HARTINGbv.nl

NO - Norway
HARTING A/S
Østensjøveien 36, N-0667 Oslo
E-Mail: no@HARTING.com, www.HARTING.no

PL - Polen
HARTING Polska Sp. z o.o.
ul. Duńska 9, Budynek DELTA, PL-54-427 Wrocław
E-Mail: pl@HARTING.com, www.HARTING.pl

PT - Portugal
HARTING Iberia, S. A.
Avda. Josep Tarradellas, 20-30, 4o 6a, E-08029 Barcelona
E-Mail: es@HARTING.com, www.HARTING.es/pt

RO - Romania
HARTING Romania SCS
Europa Unita str. 21, 550018-Sibiu, Romania
E-Mail: ro@HARTING.com, www.HARTING.com

RU - Russia
HARTING ZAO
Maily Sampsoniyevsky prospect 2A
194044 Saint Petersburg, Russia
E-Mail: ru@HARTING.com, www.HARTING.ru

SE - Sweden
HARTING AB
Gustavslundsvegen 141 B 4tr, S-167 51 Bromma
E-Mail: se@HARTING.com, www.HARTING.se

SG - Singapore
HARTING Singapore Pte Ltd.
25 International Business Park
#04-108 German Centre, Singapore 609916
E-Mail: sg@HARTING.com, www.HARTING.sg

SK - Slovakia
HARTING s.r.o.
Sales office Slovakia
J. Simora 5, SK □940 67 Nov□zky
E-Mail: sk@HARTING.com, www.HARTING.sk

TR - Turkey
HARTING TURKEI Elektronik Ltd. Sti.
Barbaros Mah. Dereboyu Cad. FesleÇen Sok.
Uphill Towers, A-1b Kat:8 D:45
34746 AtaÇehir, Çtanbul
E-Mail: tr@HARTING.com, www.HARTING.com.tr

TW - Taiwan
HARTING Taiwan Limited
Room 1, 5/F, 495 GuangFu South Road
RC-110 Taipei, Taiwan
E-Mail: tw@HARTING.com, www.HARTING.com.tw

US - USA
HARTING Inc. of North America
1370 Bowes Road, USA-Elgin, Illinois 60123
E-Mail: us@HARTING.com, www.HARTING-USA.com

ZA - South Africa
HARTING South Africa (Pty) Ltd
Ground Floor, Twickenham Building, The Campus
Cnr Main & Sloane Street Bryanston
Johannesburg (Bryanston), 2021
South Africa
E-Mail: za@HARTING.com, www.HARTING.co.za



HARTING Technologiegruppe

MarienwerderstraÙe 3 | 32339 Espelkamp
Postfach 11 33 | 32325 Espelkamp
Tel. +49 5772 47-0 | Fax +49 5772 47-400
E-Mail: de@HARTING.com | www.HARTING.com